

model-level
operators
seeded as
starting nodes

budget phase machine:
exploit until every lineage plateaus
→ explore-burst: inject 5–
8 fresh long-shot drafts
→ stop (hard cap 60 eval-
uated nodes; patience
20 evals after the burst
without a new global best)

on stop:
best node →
submission

select parent lineage
(backtrack when a lineage plateaus:
3 children without improvement)

priors from
experience library

propose mutation:
solo moves: seed / hyperparameters /
boundary push
blend moves: member set + weight search
(blends are first-class nodes
over cached OOF predictions)

duplicate gate
(SHA-256 over the canonicalized config —
params, features, seed, blend members; a
repeat burns budget, never re-evaluates)

data-level operators
each gets its own arm;
arms race, the winner
takes the full budget

evaluate in sandboxed child process
(CV score; hang or crash → SIGKILL of
the process group, node marked failed,
search continues; blend cost guard)

add node, update global best
(a solo breakthrough reopens
plateaued blend lineages)

example search tree

